Experiment 6

Q1 Answer following question in typed form and paste it in the text box provided for submission: **bold text**

1 Describe all three data set in terms of following : how many features, their categories (num,text,categorical, nominal,ordinal etc), number of tuples, missing values. **bold text**

- List item
- List item

I'll give you a complete long answer format that you can directly write in your exam/assignment. Since your datasets are not specified, I'll explain using a standard structure + example of 3 common datasets (you can replace names with your actual datasets).

Description of Three Datasets 🔹 Dataset 1: Iris Dataset **1. Number of Features** .Total Features: 4 input features + 1 target .Features: .Sepal Length .Sepal Width .Petal Length .Petal Width .Species (target)

2. Feature Categories

Feature Name	Type	Category Type
Sepal Length	Numerical	Continuous
Sepal Width	Numerical	Continuous
Petal Length	Numerical	Continuous
Petal Width	Numerical	Continuous
Species	Categorical	Nominal

Explanation:

.Numerical → measured in numbers .Nominal → no order (setosa, versicolor, virginica)

3. Number of Tuples .Total Rows: 150

4. Missing Values .Missing Values: No missing values .Dataset is clean and complete

Dataset 2: Titanic Dataset

1. Number of Features .Total Features: 12 (approx.) .Important Features: .PassengerId .Name .Age .Sex .Ticket .Fare .Cabin .Embarked .Survived (target)

2. Feature Categories

Scanned with OKEN Scanner

Feature Name	Type	Category Type
Age	Numerical	Continuous
Fare	Numerical	Continuous
PassengerId	Numerical	Discrete
Sex	Categorical	Nominal
Embarked	Categorical	Nominal
Pclass	Categorical	Ordinal
Name	Text	-
Cabin	Text	-

3. Number of Tuples .Total Rows: 891

4. Missing Values .Age → Missing values present .Cabin → Many missing values .Embarked → Few missing values

Dataset 3: Marathon Dataset

1. Number of Features .Total Features: 5–6 (approx.) .Features: .Name .Age .Gender .Split Time .Final Time .Pace

2. Feature Categories

Feature Name	Type	Category Type
Age	Numerical	Discrete
Split Time	Numerical	Continuous
Final Time	Numerical	Continuous
Pace	Numerical	Continuous
Gender	Categorical	Nominal
Name	Text	-

3. Number of Tuples .Total Rows: ~37,000 (approx.)

Double-click (or enter) to edit

4. Missing Values .Some missing values may exist in: .Age .Split/Final Time .Needs cleaning and preprocessing

Overall Conclusion Iris Dataset .Small, clean dataset .No missing values .Mostly numerical features **Titanic Dataset** .Mixed data types (numerical, categorical, text) .Contains missing values .Requires preprocessing **Marathon Dataset** .Large dataset .Mostly numerical + some categorical .May contain missing values

2 In IRIS data set which non graphical EDA is performed ?

Scanned with OKEN Scanner

Non-Graphical EDA performed on Iris Dataset

Non-graphical EDA means analyzing data using numbers, statistics, and summaries (without plots).

1. Basic Information .Check number of rows and columns .Check data types of each feature

Example:

.Total rows: 150 .Total features: 5 (4 numerical + 1 categorical)

2. Summary Statistics .Calculate statistical values for numerical features: .Mean .Median .Standard Deviation .Minimum & Maximum .Quartiles (Q1, Q2, Q3)

Helps to understand distribution of:

.Sepal Length .Sepal Width .Petal Length .Petal Width

3. Missing Values Check Identify null or missing values

Result: **bold text**

No missing values in Iris dataset

4. Unique Values / Frequency Count Check unique values in categorical feature (Species)

** Example:**

.Setosa = 50 .Versicolor = 50 .Virginica = 50

5. Correlation Analysis .Measure relationship between numerical features

Example:

.Petal Length and Petal Width → highly correlated .Sepal features → less correlated

6. Data Distribution (Numerical Check) .Analyze spread using: .Range .Variance

7. Outlier Detection (Numerical Method) .Using statistical methods like: .IQR (Interquartile Range) .Z-score

3 In Air Quality data set which graph is used to perform multivariate analysis?

Multivariate Analysis in Air Quality Dataset

For multivariate analysis (analyzing more than two variables together) in the Air Quality dataset, the most commonly used graph is:

Scanned with OKEN Scanner

Pair Plot A pair plot shows relationships between multiple numerical variables at the same time It displays: Scatter plots between every pair of variables Distribution (histogram) on the diagonal ** Why it is used:** Helps to understand correlation between pollutants like: CO NO_2 O_3 Temperature Humidity Detects: Relationships Patterns Outliers

Other Graphs for Multivariate Analysis 1. Heatmap Shows correlation matrix Easy to identify strong/weak relationships **2. 3D Scatter Plot** Used when analyzing 3 variables together

4 bold text In Wine quality data set can you do non graphical EDA ? Explain your answer.

Non-Graphical EDA on Wine Quality Dataset

Yes, non-graphical EDA can be performed on the Wine Quality dataset, and it is very useful to understand the data using statistical summaries instead of plots.

1. Dataset Overview The dataset contains physicochemical properties of wine such as: Fixed acidity Volatile acidity Citric acid Residual sugar Chlorides Alcohol Quality (target)

Most features are numerical, and quality is:

Ordinal categorical (values like 3–8, ordered ratings)

2. Non-Graphical EDA Techniques ** (a) Basic Information** Check: Number of rows and columns Data types (mostly numerical)

Helps understand dataset structure

(b) Summary Statistics Calculate: Mean Median Standard deviation Min & Max Quartiles

Helps to understand:

Central tendency Spread of each feature

(c) Missing Values Analysis Check for null values

Result:

Wine dataset usually has no missing values

(d) Frequency Count (Quality) Count number of samples for each quality level

** Example:**

Quality 5 → most common Quality 3 or 8 → rare

Shows dataset imbalance

(e) Correlation Analysis Compute correlation between variables

Scanned with OKEN Scanner

Example:

Alcohol → positively correlated with quality Volatile acidity → negatively correlated with quality

(f) Outlier Detection (Statistical) Using: IQR (Interquartile Range) Z-score

Detect extreme values in:

Alcohol Residual sugar

(g) Range & Variance Measure spread of values for each feature

Explanation (Why Non-Graphical EDA is Suitable) The Wine dataset is numerical-heavy, so statistical analysis is effective It helps to: Understand data distribution Identify relationships Detect anomalies It is useful when visualization is not required or not possible

Close

```
**Conclusion**

Yes, non-graphical EDA is effectively perfor
using summary statistics, correlation analys
value checks. These methods provide a clear
relationships, and quality distribution with
```

Conclusion

Yes, non-graphical EDA is effectively performed on the Wine Quality dataset using summary statistics, correlation analysis, frequency counts, and missing value checks. These methods provide a clear understanding of data structure, relationships, and quality distribution without using graphs.

Scanned with OKEN Scanner